

Parametric Investigation of the Operating Limits and Performance Effects of Anode Off-Gas Recirculation in Ammonia-Fueled Solid Oxide Fuel Cell Systems

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Supplementary Table S1 Stream data of system base case operation.

Stream number	T [°C]	p [atm]	$\dot{m}$ [kg/h]	Chemical composition [mol fraction %]				
				H ₂	H ₂ O	O ₂	N ₂	NH ₃
1	45.0	4.221	14.81	0	0	0	0	1
2	456.8	1.224	46.82	0.08823	0.41176	0	0.16666	0.33334
3	602.1	1.137	46.82	0.44092	0.3089	0	0.24994	0.00025
4	699.9	1.130	64.01	0.13235	0.61765	0	0.25	0
5	699.9	1.130	32.00	0.13235	0.61765	0	0.25	0
6	699.9	1.130	32.00	0.13235	0.61765	0	0.25	0
7	25.0	1.000	662.40	0	0	0.21	0.79	0
8	38.0	1.136	662.40	0	0	0.21	0.79	0
9	599.2	1.135	662.40	0	0	0.21	0.79	0
10	699.9	1.130	645.19	0	0	0.19107	0.80893	0
11	699.9	1.130	396.79	0	0	0.19107	0.80893	0
12	699.9	1.130	248.40	0	0	0.19107	0.80893	0
13	474.1	1.036	248.40	0	0	0.19107	0.80893	0
14	692.6	1.036	677.20	0	0.054257	0.17338	0.77237	0
15	692.6	1.036	561.42	0	0.054257	0.17338	0.77237	0
16	692.6	1.036	115.80	0	0.054257	0.17338	0.77237	0
17	59.6	1.000	561.42	0	0.054257	0.17338	0.77237	0
18	173.4	1.000	677.20	0	0.054257	0.17338	0.77237	0

Supplementary Table S2 Stream data of ejector back pressure – entrainment stream temperature dependence analysis.

(Stream properties of both flows refer to the ejector primary and entrained flow of base case operation.)

Stream Data							
Stream	T [°C]	p [atm]	$\dot{m}$ [kg/h]	Chemical composition [mol fraction %]			
				H ₂	H ₂ O	N ₂	NH ₃
Primary flow	45	4.221	14.81	0	0	0	1
Entrained flow	Varies	1.130	32.00	0.13235	0.61765	0.25	0

Supplementary Table S3 Summary of ejector back pressure – entrainment stream temperature dependence analysis.

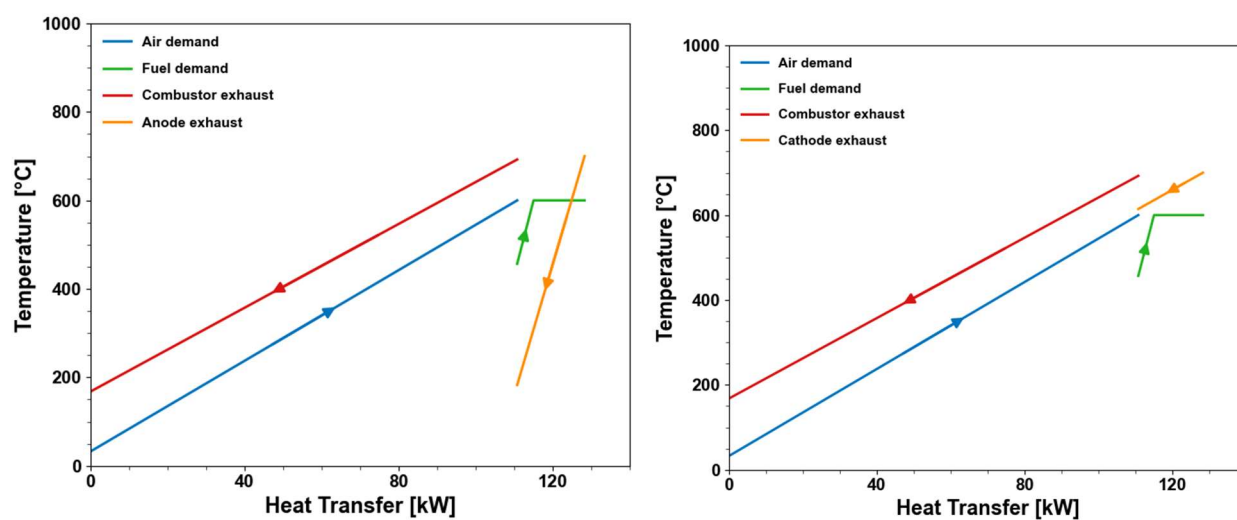
Entrained flow temperature [°C]	Mixed flow Mach number	Ejector back pressure [atm]
700	1.072	1.224
750	1.068	1.218
800	1.063	1.212
850	1.059	1.206
900	1.055	1.200

Supplementary Table S4 Summary of fuel cracker heat duty, LMTD and hot flow mass flowrate at different AOCR ratio.(System operation under base case $F_{U_{stack}}$, j , T_{stack} .)

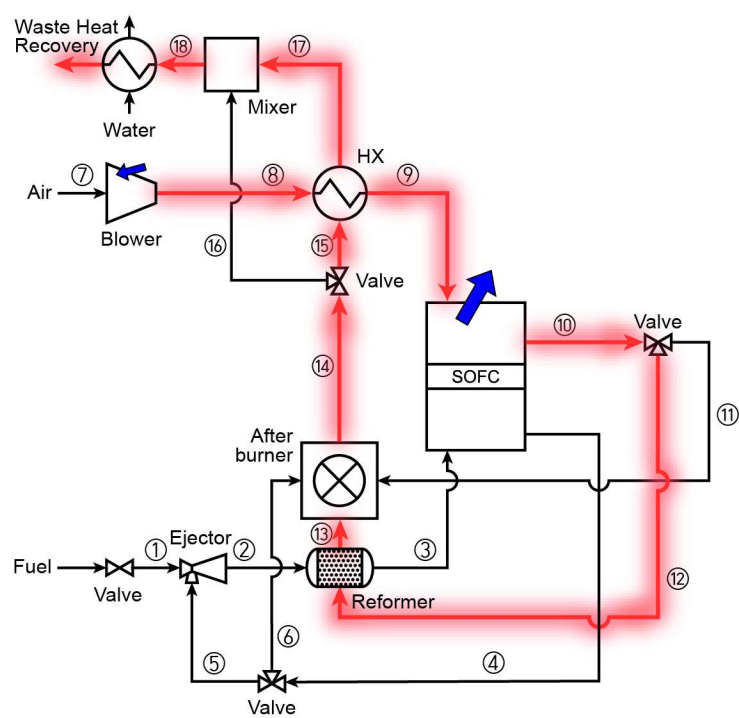
AOCR ratio [%]	Reformer heat duty [kW]	LMTD [°C]	Required mass flow rate of cracker hot flow [kg/h]
0	22.58	100.24	135.34
10	21.69	84.12	150.61
20	20.80	72.29	168.13
30	19.80	63.44	189.92
40	18.70	54.88	215.44
50	17.50	47.14	248.40
60	15.99	39.29	290.45
66.6	14.75	10.12	311.53

Supplementary Table S5 Summary of fuel cracker heat duty, LMTD and hot flow mass flowrate at different AOCR ratio.(System operation under base case $F_{U_{stack}}$, j , T_{stack} .)

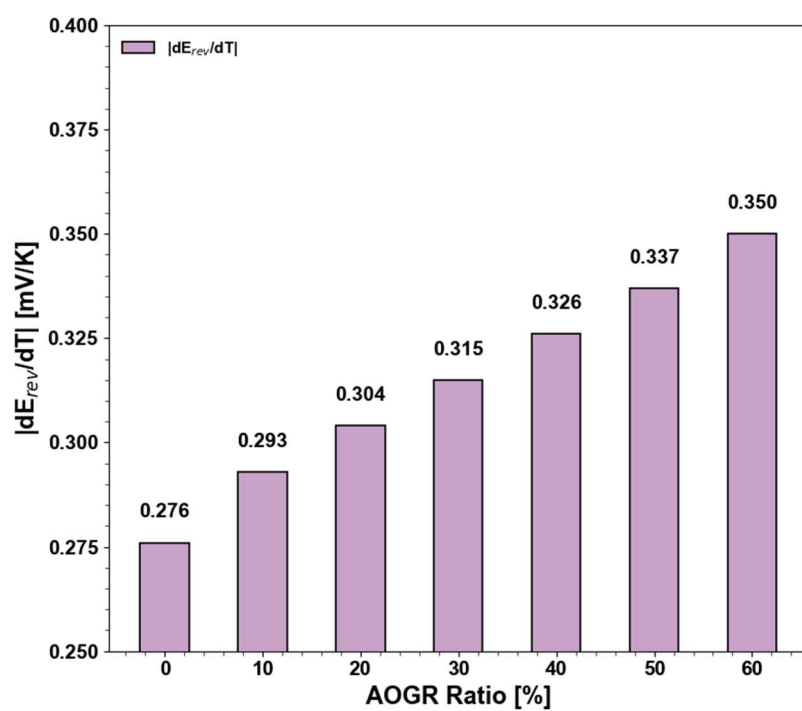
		T_{stack} [°C]				
		600	650	700	750	800
Reversible potential		0.9688	0.9521	0.9353	0.9184	0.9013
Overpotential	Activation	0.0520	0.0322	0.0197	0.0124	0.0081
	Ohmic	0.0507	0.02699	0.0152	0.0090	0.0056
	Concentration	0.0024	0.0025	0.0025	0.0025	0.0026
	Total	0.1051	0.0617	0.0375	0.0240	0.0163
Unit cell voltage		0.8637	0.8905	0.8978	0.8944	0.8850



Supplementary Fig. S1. T-Q diagrams of systems utilizing reformer heat source from: (a) anode off-gas; (b) cathode off-gas.



Supplementary Fig. S2. Flow path of fluid to be pressurized by air blower in a recirculation system.



Supplementary Fig. S3. The correlation between AOGR ratio and magnitude of E_{rev} drop sensitivity to T_{stack} .